

THEORY



Training Program 01

Advanced NLP & Large Language Models (LLMs)



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Training Program 01

M01: Text Preprocessing

M02: Word and Sentence Embeddings

M03: Transformer Architectures

M04: Entity-Centric NLP

M05: Applied NLP Tasks

M06: Multimodal NLP

M07: LLM Fundamentals

M08: Prompt Engineering

M09: Fine-Tuning & PEFT

M10: RAG & Real-World Projects



MODULE M01

TEXT PREPROCESSING

CHAPTER M02

Embeddings and Vector Representations



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1. Why Vectors? (Foundations of Embeddings)

2. Word2Vec (CBOW & Skip-gram)
3. GloVe (Global Co-occurrence Models)
4. fastText (Subword-aware Static Embeddings)
5. BERT and Contextual Embeddings
6. Sentence Embeddings (SBERT & Semantic Search)
7. Synthesis and Transition to Transformers



Chapter 2.1 Introduction to Embeddings

- Embeddings map discrete symbols into numerical vectors.
- Words and tokens become points in a continuous space.
- Similar meanings are represented by nearby vectors.
- This enables efficient mathematical processing of language.



Chapter 2.1 Introduction to Embeddings

The Distributional Hypothesis

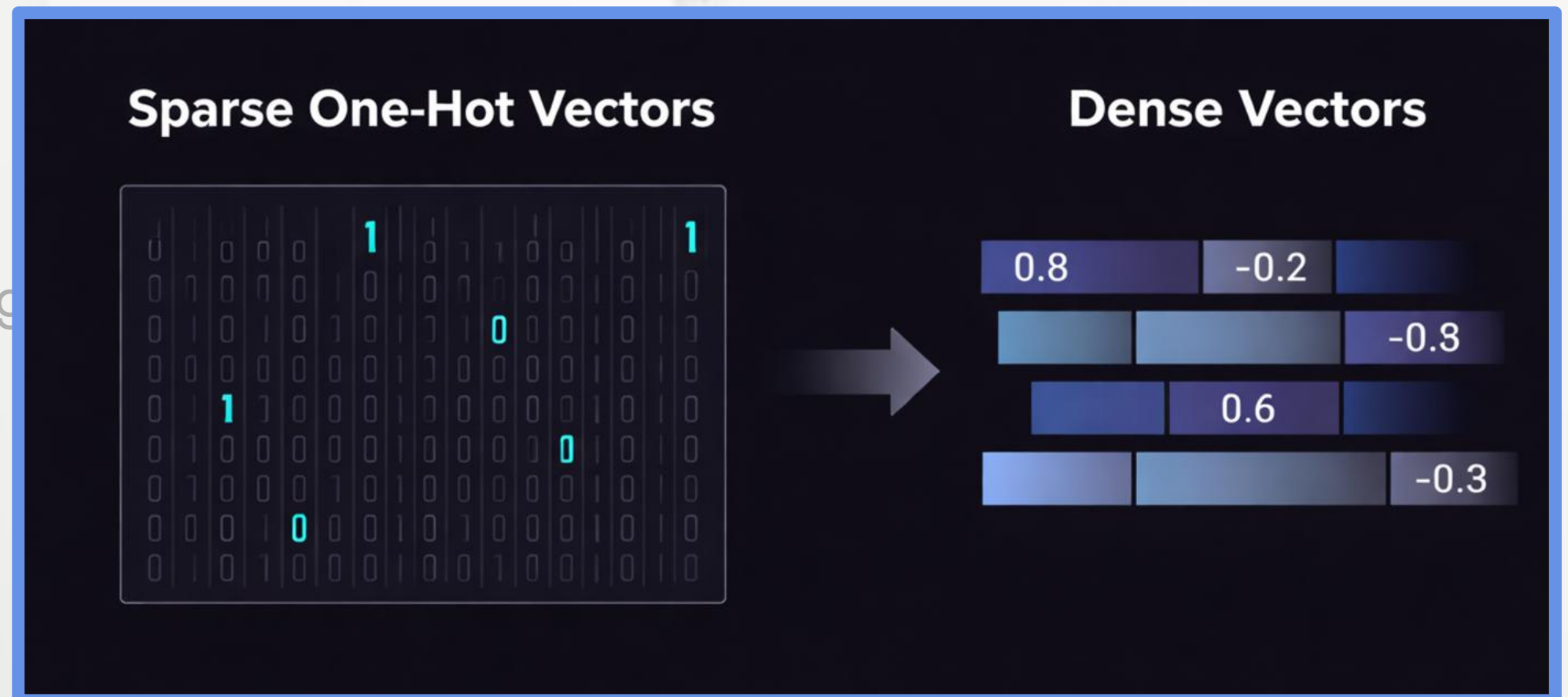
- Meaning emerges from usage and context.
- Words appearing in similar contexts share meaning.
- Models learn semantics without explicit definitions.
- Context statistics drive semantic understanding.



Chapter 2.1 Introduction to Embeddings

From One-Hot Encoding to Dense Vectors

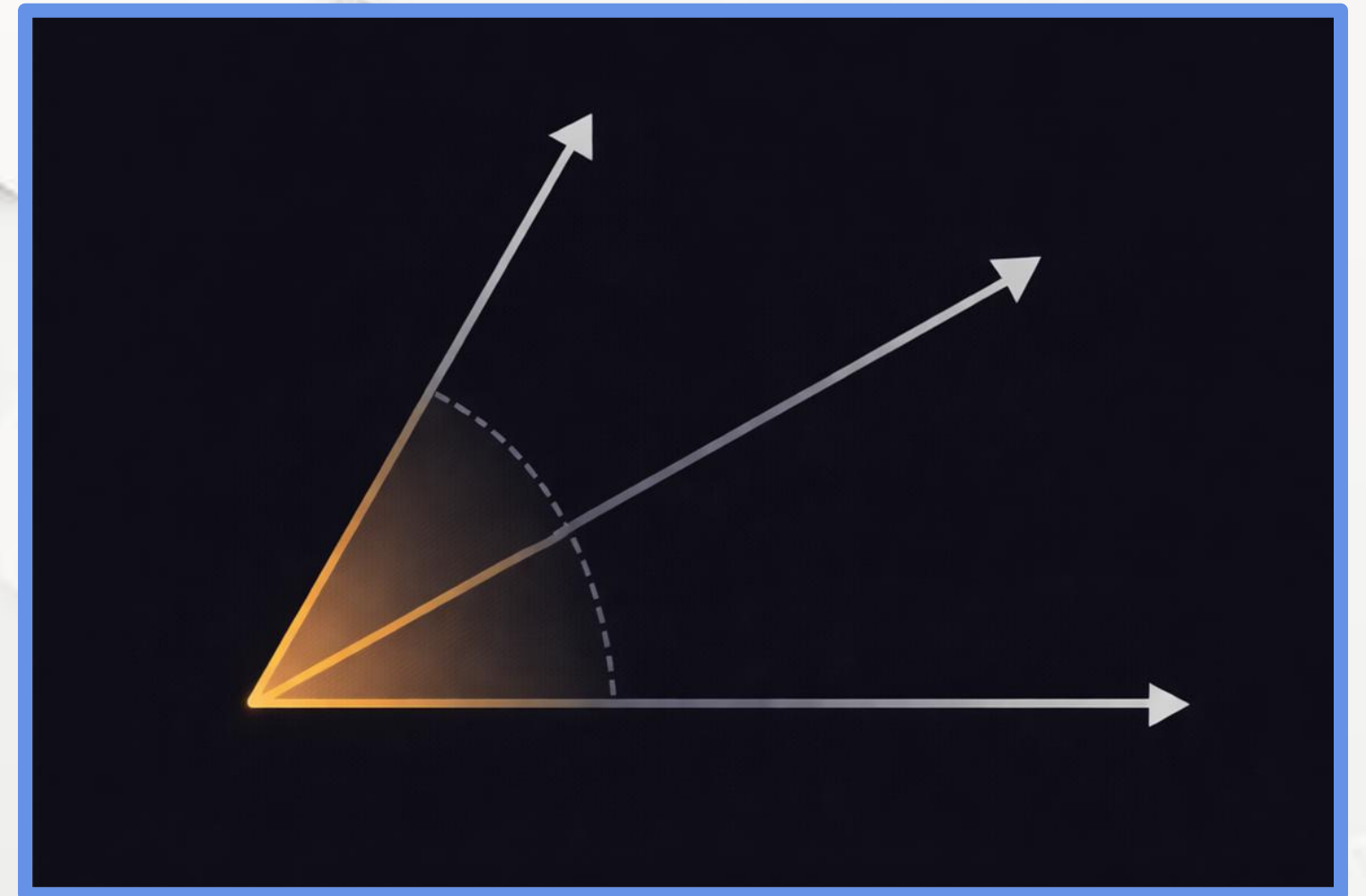
- One-hot vectors are sparse and extremely high dimensional.
- All words are orthogonal and unrelated.
- Dense embeddings reduce dimensionality.
- Semantic similarity becomes measurable.



Chapter 2.1 Introduction to Embeddings

The Distributional Hypothesis

- Vector similarity measures semantic closeness.
- Cosine similarity compares vector orientation.
- Similar meanings produce high similarity scores.
- This enables search, clustering, and retrieval.



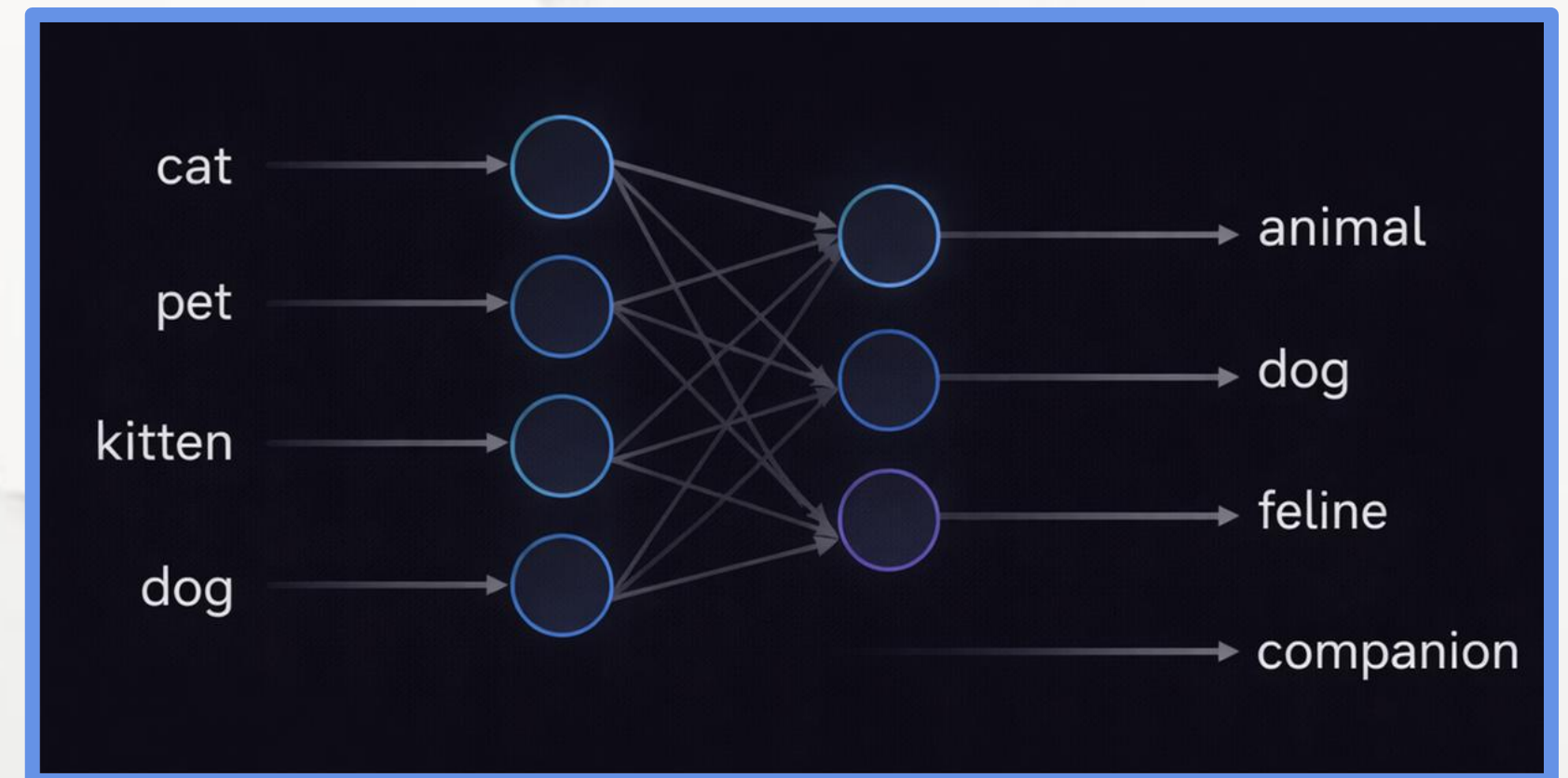
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Chapter 2.2 Word2Vec

What Is Word2Vec?

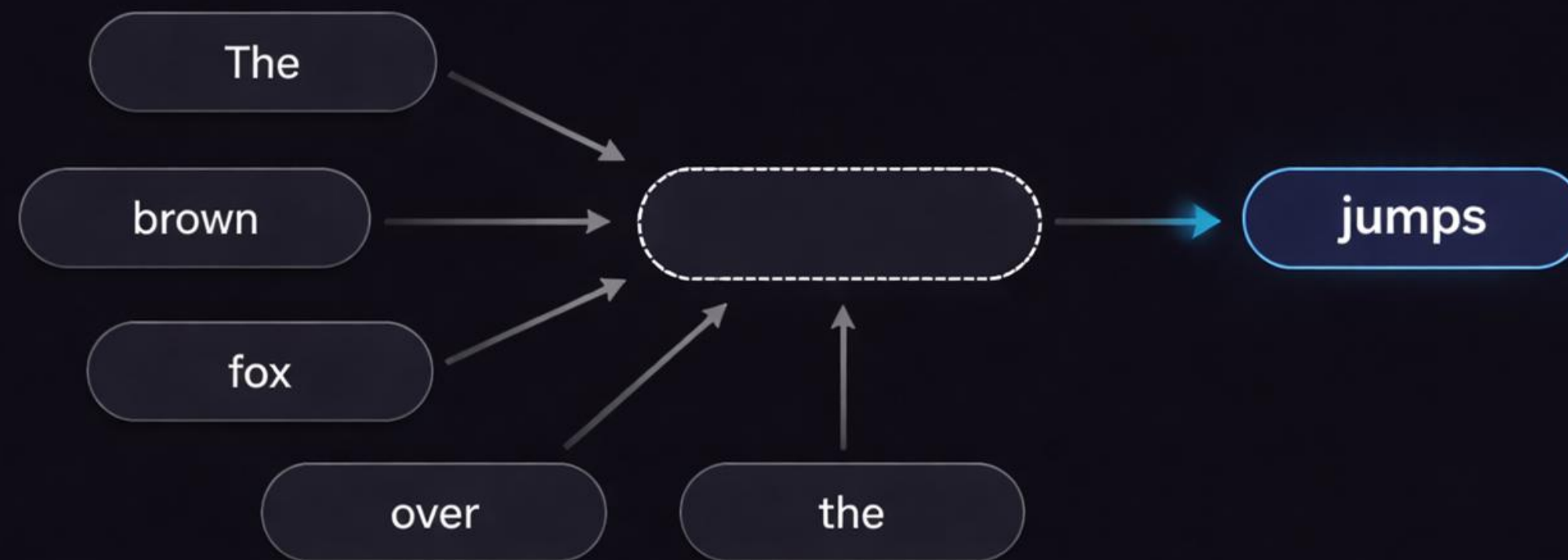
- Word2Vec learns word meanings from raw text.
- It uses self-supervised learning from context.
- Words are embedded into a shared vector space.
- Meaning emerges from prediction tasks.



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Chapter 2.2 Word2Vec

Continuous Bag-of-Words (CBOW)

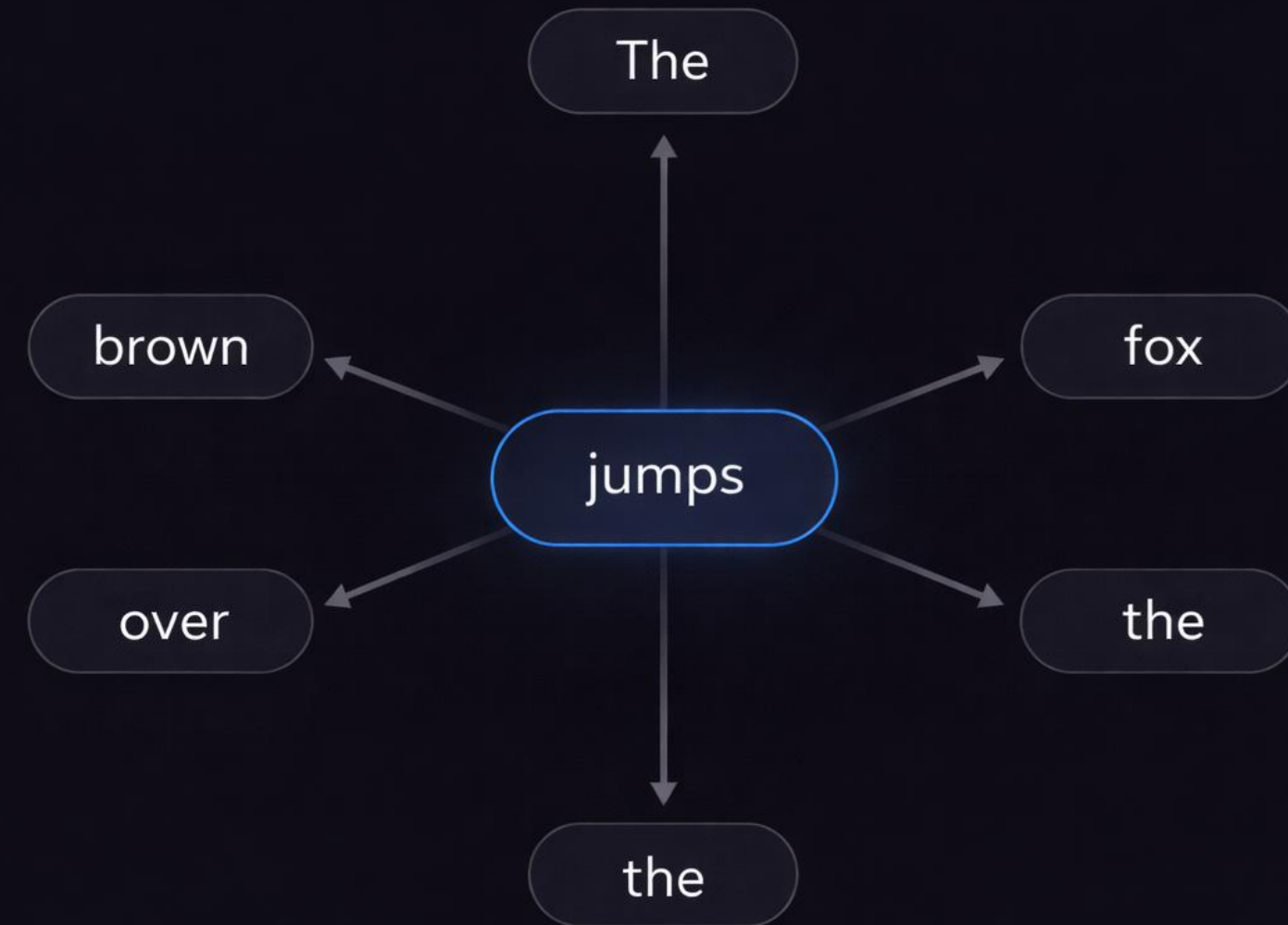


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Chapter 2.2 Word2Vec

Skip-gram Model



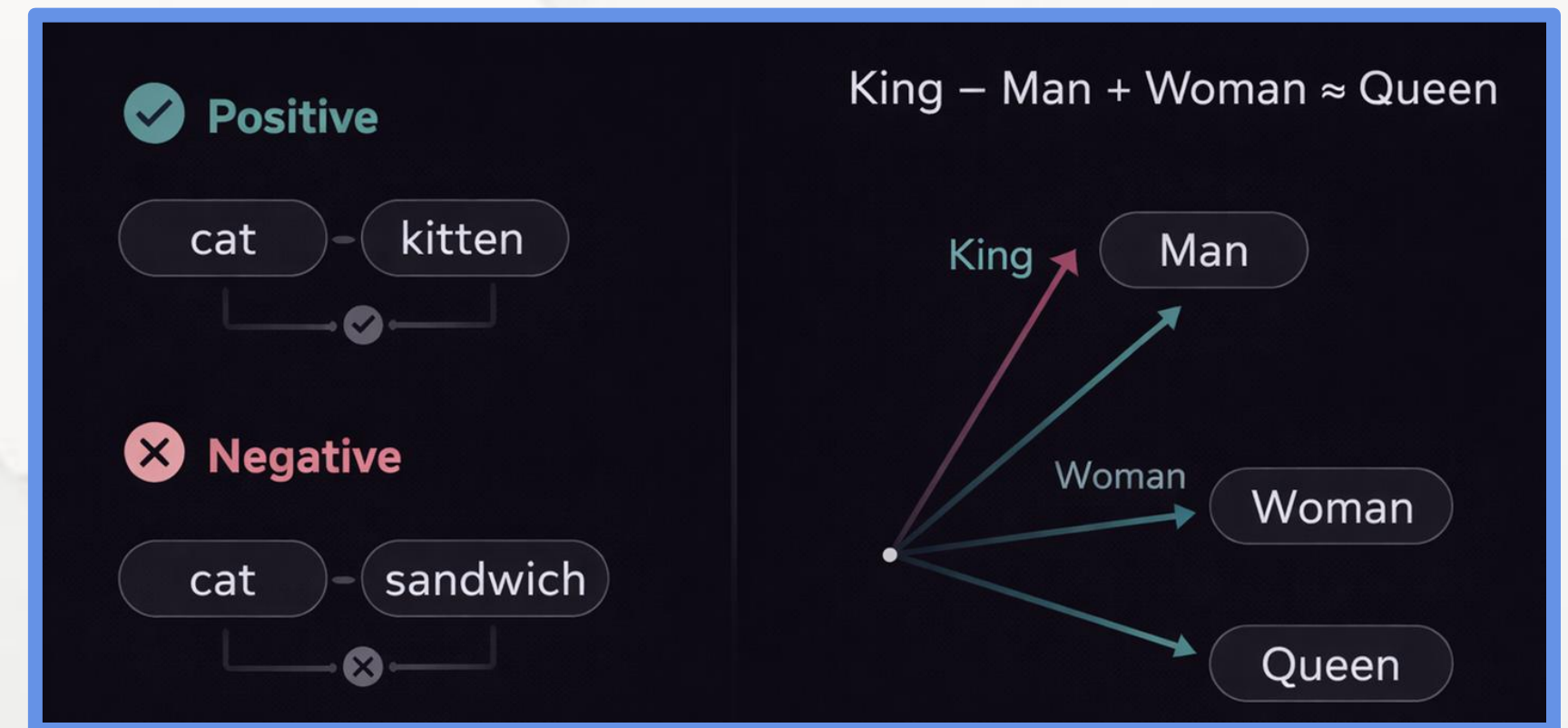
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Chapter 2.2 Word2Vec

Negative Sampling and Word Analogies

- Full softmax is computationally expensive.
- Negative sampling simplifies training.
- Only a few negative examples are updated.
- Word vectors capture linear semantic relations.



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Chapter 2.3 GloVe: Global Vectors for Word Representation

Why GloVe Co-occurrence Matters

- Word2Vec learns from local context windows.
- Some semantic patterns are global, not local.
- Global co-occurrence captures corpus-wide statistics.
- GloVe combines local and global information.



Chapter 2.3 GloVe: Global Vectors for Word Representation

The GloVe Co-occurrence Matrix

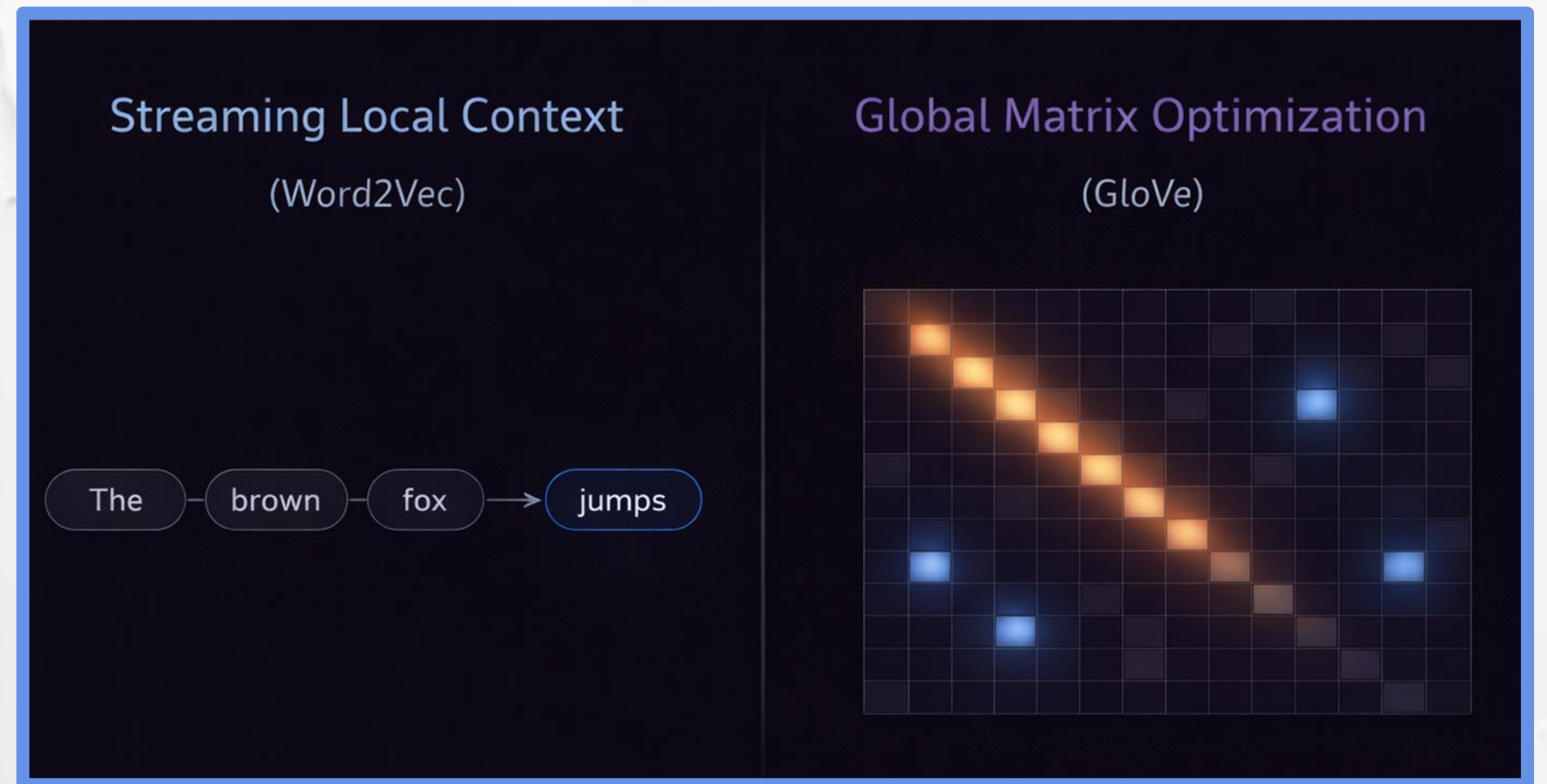
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- GloVe combines local and global information.



Chapter 2.3 GloVe: Global Vectors for Word Representation

GloVe vs Word2Vec

- Word2Vec is prediction-based and local.
- GloVe is count-based and global.
- Both produce dense word embeddings.
- GloVe often excels in similarity and analogy tasks.



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Chapter 2.4 fastText

Limitations of Static Word Embeddings

- Word2Vec and GloVe assign one vector per word.
- Unseen words are mapped to an unknown token.
- Morphological variations are poorly represented.
- This limits robustness in real-world text.

The brown fox **jumps** over the log.

<UNK>

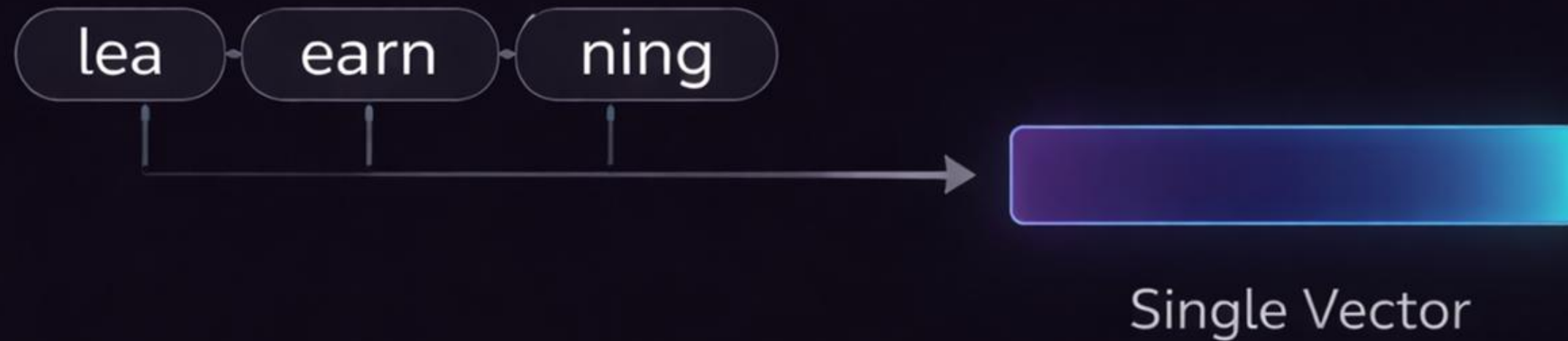


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Chapter 2.4 fastText

Character N-grams in fastText

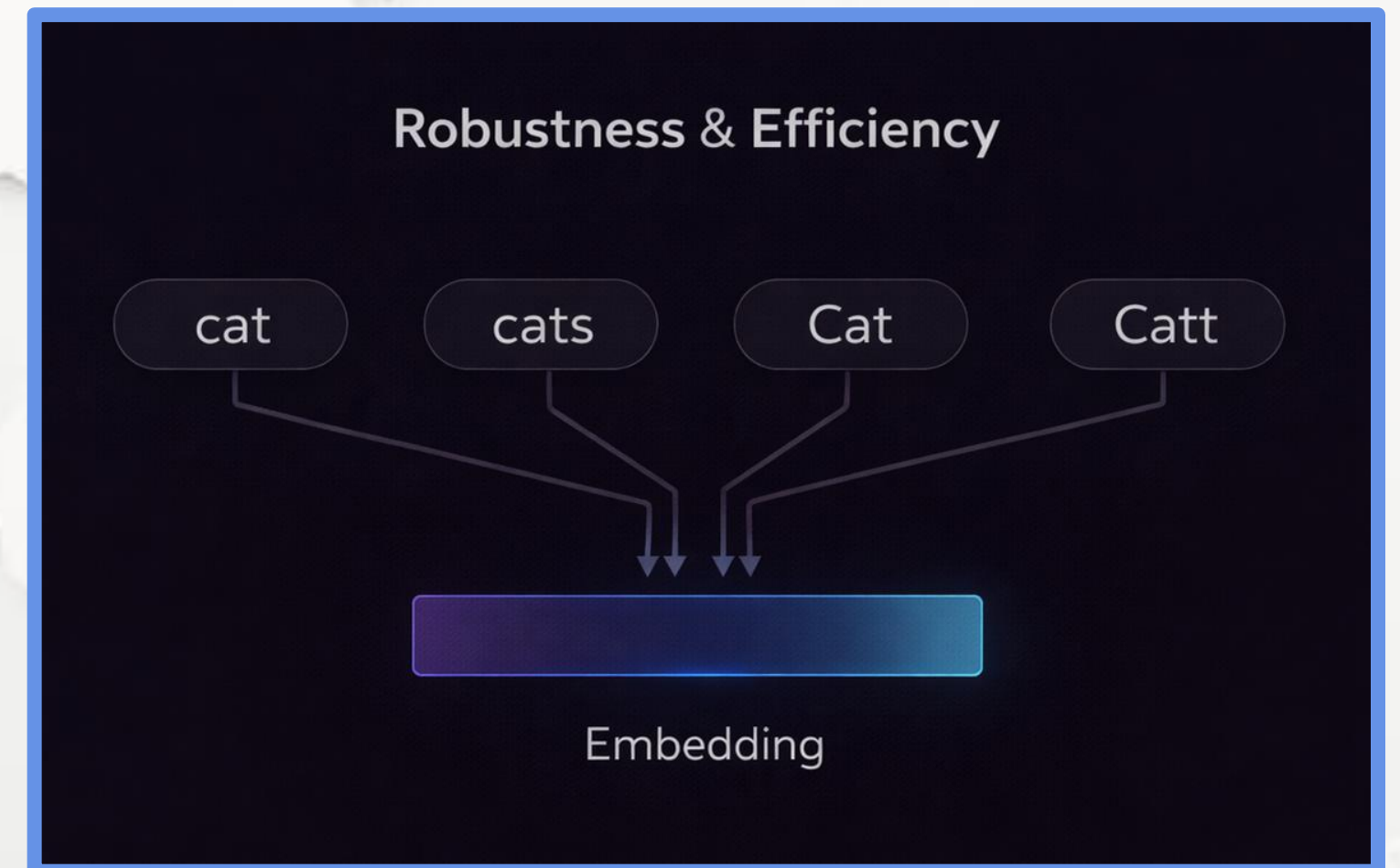


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Chapter 2.4 fastText

Practical Advantages of fastText

- Strong performance on morphologically rich languages.
- Robust to typos and spelling variations.
- Fast training and low inference cost.
- Widely used for large-scale text classification.



Training Program 01

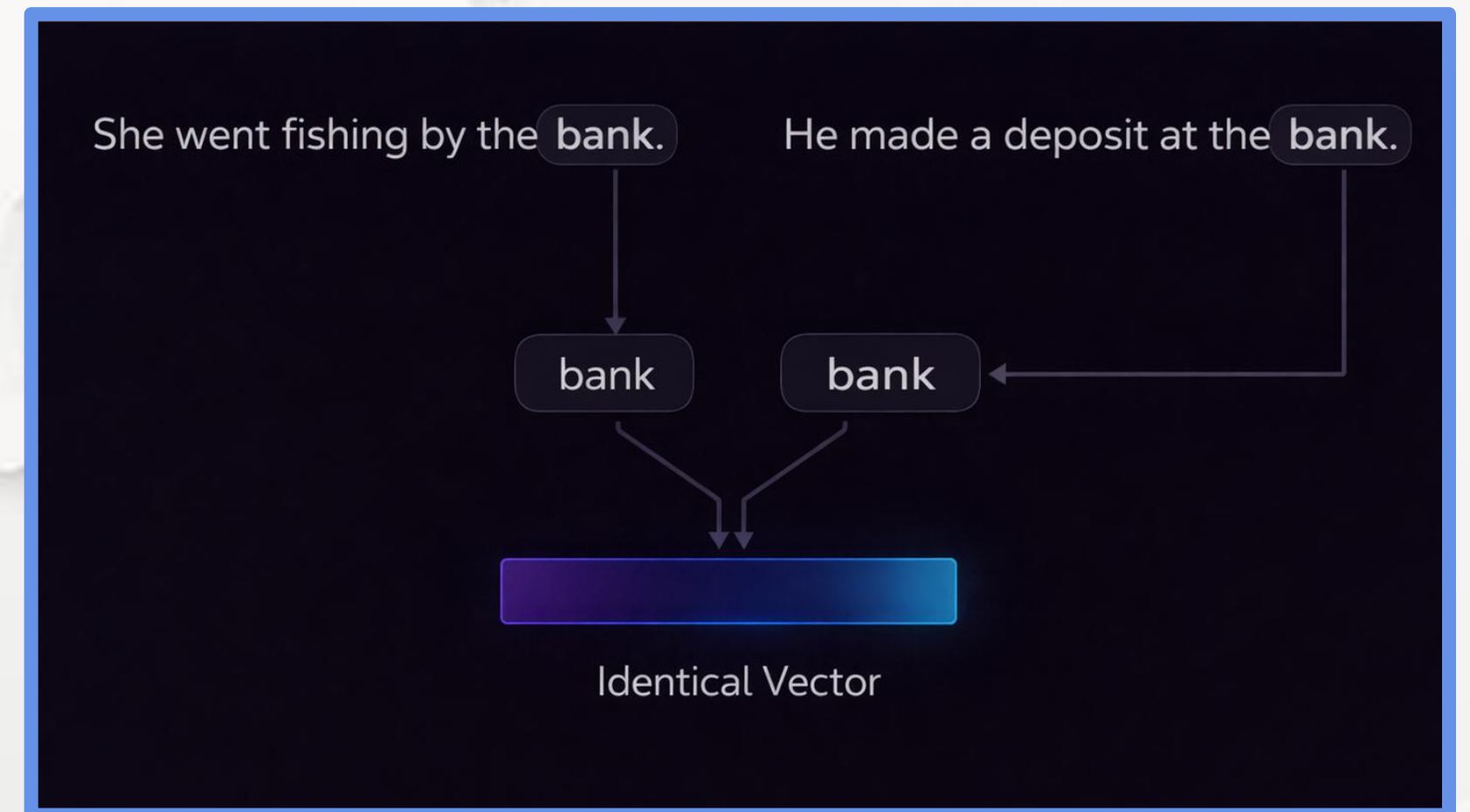
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Chapter 2.5 BERT

Practical Advantages of fastText

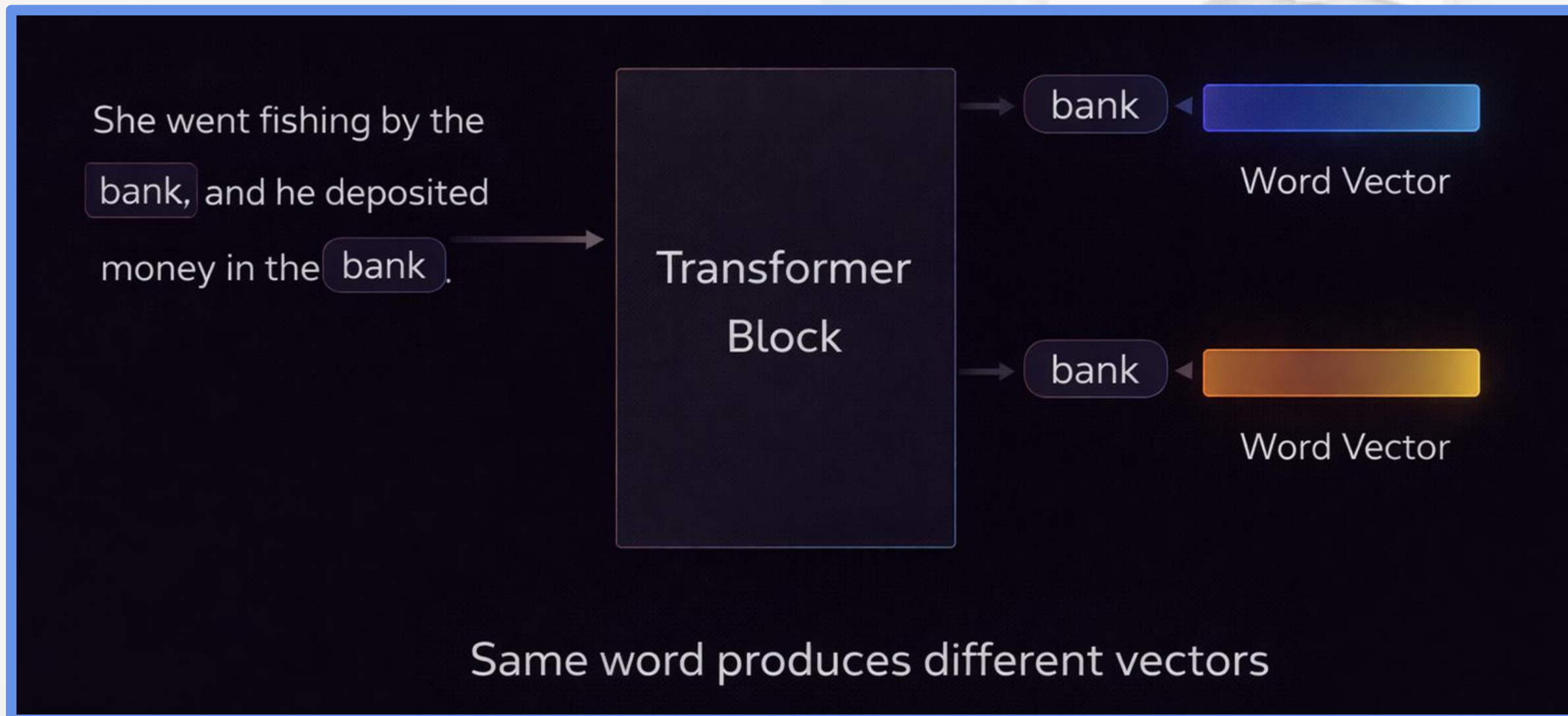
- Static embeddings assign one vector per word.
- Meaning does not change with context.
- Polysemous words share a single representation.
- This limits language understanding.



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Chapter 2.5 BERT

Contextual Embeddings with Transformers

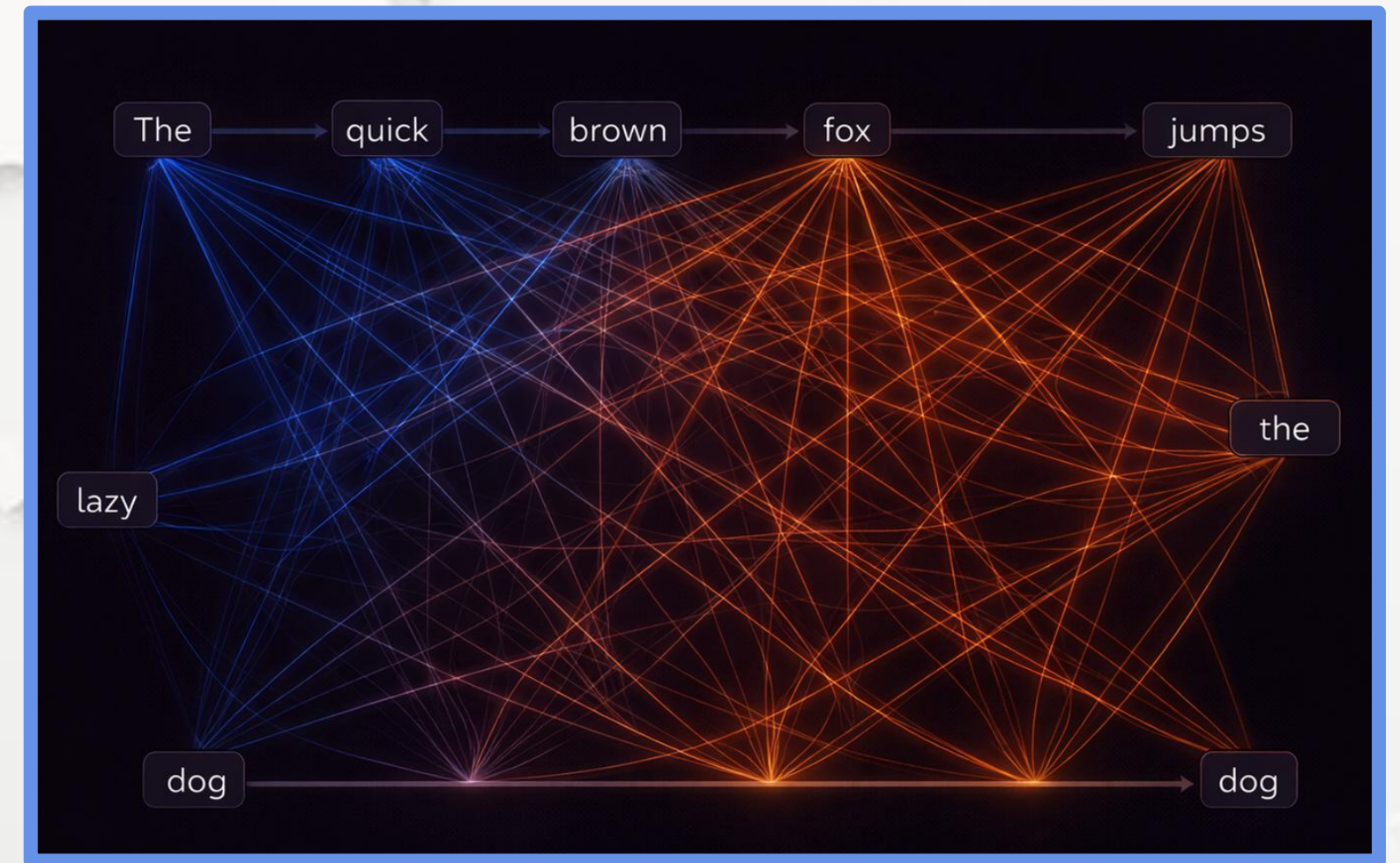


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Chapter 2.5 BERT

Bidirectional Self-Attention

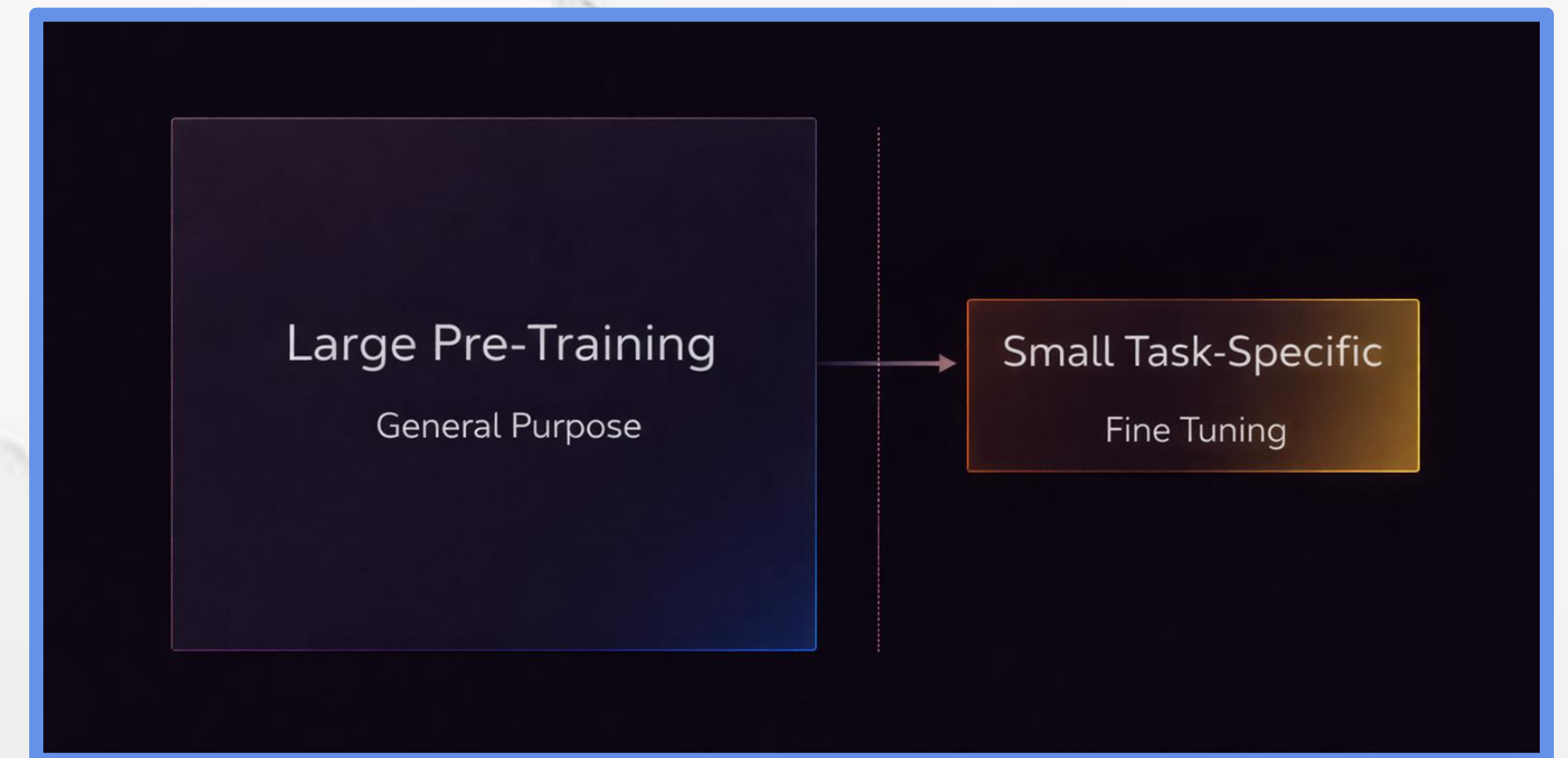
- BERT processes text in both directions.
- Each token attends to all other tokens.
- Context is built globally, not sequentially.
- This enables deeper semantic understanding.



Chapter 2.5 BERT

Pre-training and Fine-tuning

- Static embeddings assign one vector per word.
- Meaning does not change with context.
- Polysemous words share a single representation.
- This limits language understanding.



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Chapter 2.6 Sentence Embeddings

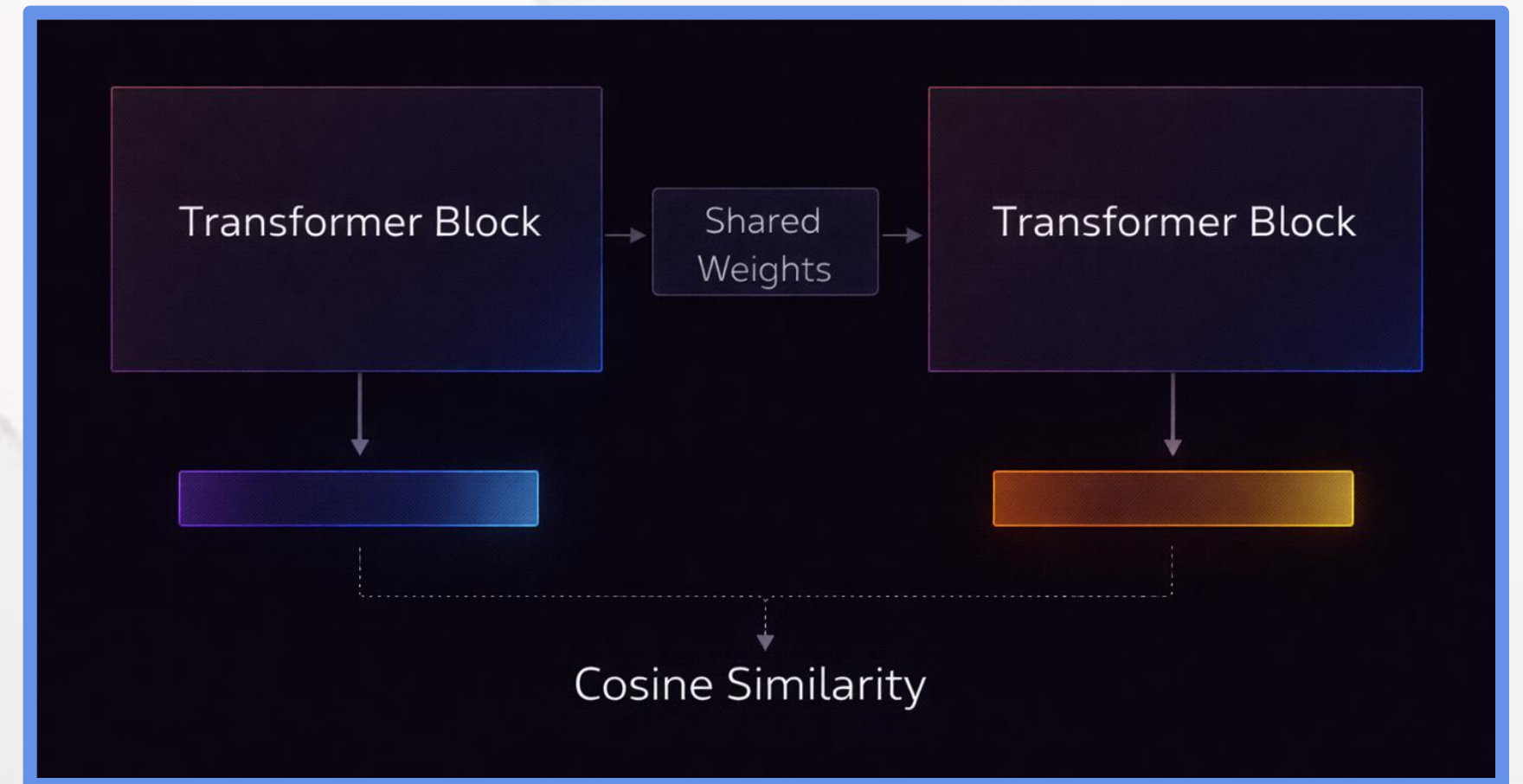
Why Token-Level Embeddings Are Not Enough



Chapter 2.6 Sentence Embeddings

Sentence-BERT (SBERT) Architecture

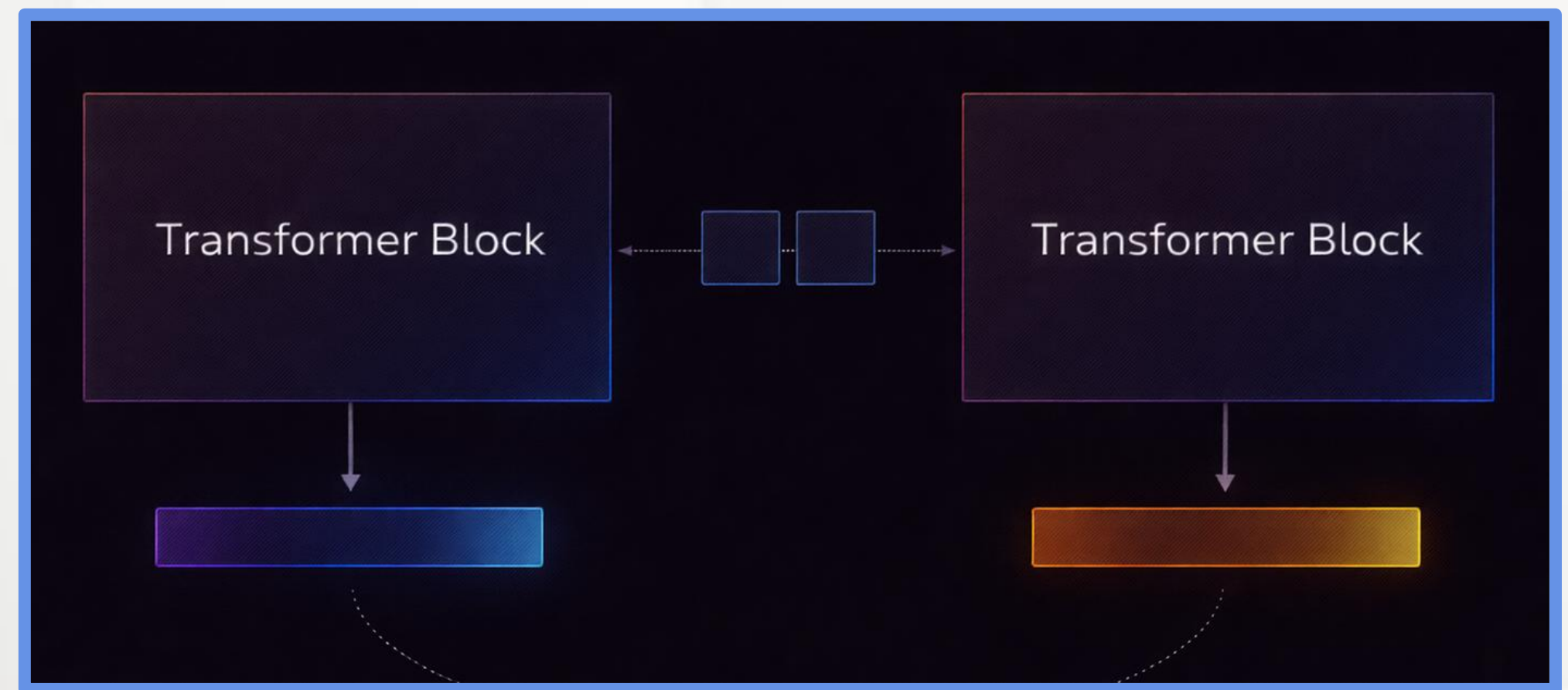
- SBERT uses a Siamese transformer architecture.
- Sentences are encoded independently.
- Vector comparison uses cosine similarity.
- Similarity computation is fast and scalable.



Chapter 2.6 Sentence Embeddings

Sentence-BERT (SBERT) Architecture

- Semantic search over large document collections.
- Retrieval-Augmented Generation (RAG).
- Clustering and topic discovery.
- Near-duplicate detection.



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Chapter 2.7 Synthesis of Embedding Methods

From Static to Contextual Embeddings

- Static embeddings map words to fixed vectors.
- Subword methods improve robustness and coverage.
- Contextual models adapt meaning to usage.
- Sentence embeddings enable scalable semantics.

Increasing Semantic Richness

Word

Subword

Contextual

Sentence

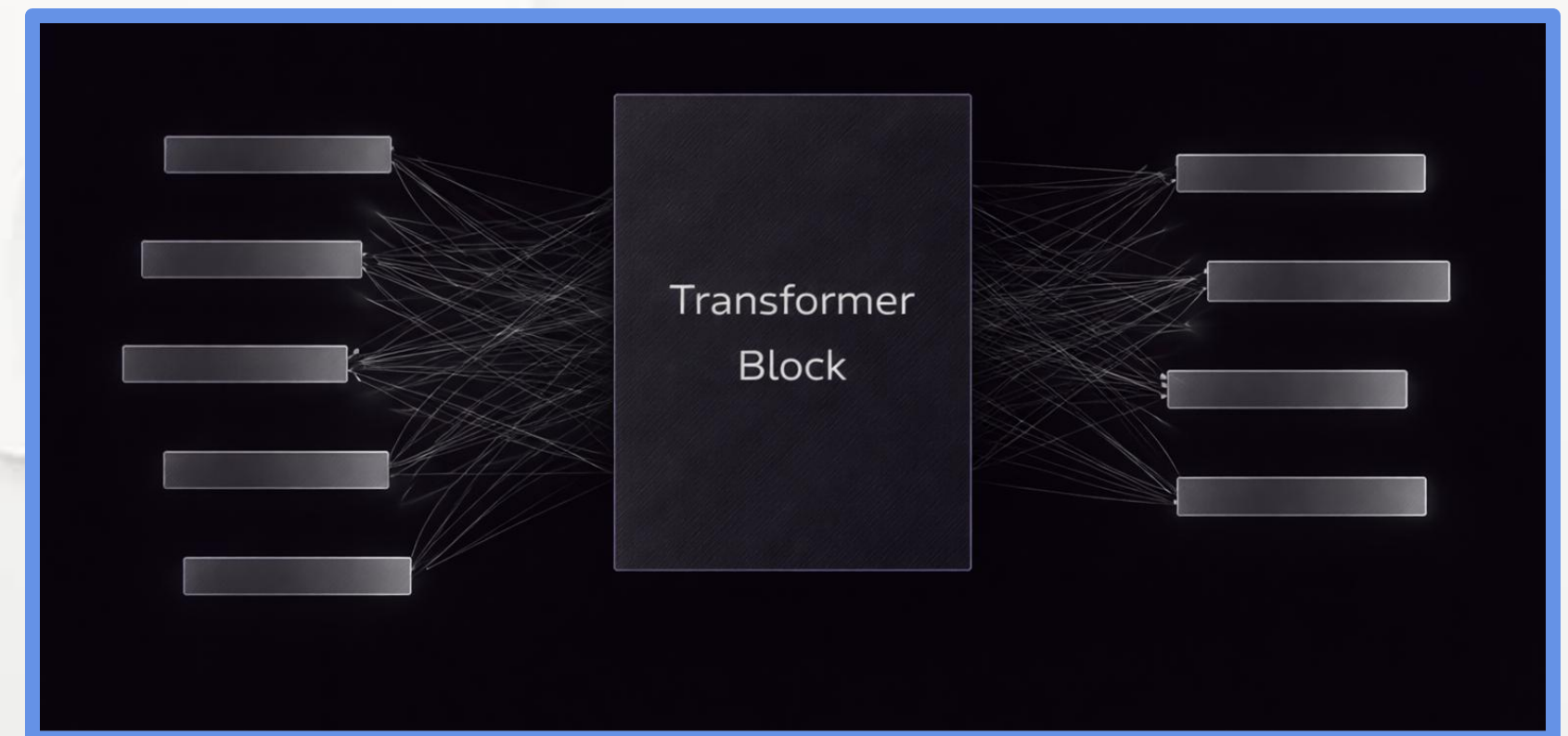


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Chapter 2.7 Synthesis of Embedding Methods

Why Transformers Are the Next Step

- Embeddings alone do not perform reasoning.
- Transformers process vectors through attention.
- Meaning emerges from interaction, not isolation.
Architecture defines intelligence.





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